

of threads, AI cores are designed for hyperscale computing and can handle thousands or even millions of threads simultaneously.

What are the design challenges in making an AI core processor that can operate on battery power?

With traditional architectures, such as GPUs or x86 processors, running thousands of computations in parallel requires thousands of cores or multiple chips. This is highly inefficient because these systems were not designed for large-scale parallelism. The primary challenge is power consumption. Running thousands of computations on traditional digital systems can drain a battery in seconds or minutes. To create a device capable of operating for weeks or months, we required an architecture that could efficiently handle massive parallel computing.

Our solution is analogue AI computing, where core operations such as multiply-and-accumulate are performed using analogue circuits. Analogue AI is highly efficient but introduces challenges, including reliability issues and sensitivity to variations. To address this, we developed a hybrid architecture combining the strengths of digital and analogue computing. We call this the DigAn architecture, derived from digital and analogue. It retains over 80 per cent of the efficiency of analogue computing while eliminating unreliable behaviour.

Are there any trade-offs with accuracy when designing such a low-power AI core?

In a purely analogue AI architecture, as adopted by some companies since around 2012 or 2013, there is typically a trade-off between power efficiency and accuracy. Some organisations accept reduced accuracy to achieve power savings. In our case, however, we chose not to compromise on accuracy.

We modified the analogue design

and made selective trade-offs in power efficiency to maintain the same level of accuracy as a digital computer. As a result, our DigAn architecture computes accurately without the errors or uncertainty inherent in pure analogue AI systems.

What are the advantages of integrating analogue compute blocks for AI workloads?

The primary advantage is efficiency. A digital AI implementation typically requires 300 to 500 transistors for a given operation, whereas an analogue approach can perform the same operation using only a few transistors. In our design, we slightly increased the transistor count compared to a pure analogue implementation, resulting in a 20-30 per cent increase in area and power.

Despite this increase, the power savings from analogue computation remain one to two orders of magnitude higher than those of a fully digital design.

You mentioned a 20 to 30 per cent increase in area. Is this compared to classical analogue designs?

Classical analogue designs are at least two orders of magnitude more efficient than digital designs in terms of area and power because they trade accuracy for efficiency, accepting some level of error. In our case, we prioritised accuracy and chose not to compromise, which makes our solution approximately 20-30 per cent less efficient than a classical analogue design.

What challenges arise when mapping digital neural network models onto analogue compute elements?

If computational errors were present, this would create significant challenges. However, we have developed a block that accepts digital input, performs analogue computation, and produces digital output.

The conversion from digital to analogue and back is fully accurate, so from an application developer's perspective, there is no difference between traditional digital computing and our DigAn architecture.

This is the core of our innovation. We provide the same flexibility, accuracy, and software paradigm and toolchain that developers already use, while delivering extremely low power consumption and high performance.

How does ADC design impact system performance in a mixed-signal AI processor?

ADC design is critical because it directly affects the accuracy of signal conversion between analogue and digital domains. Our team has introduced significant innovations by converting certain portions of the ADC into partially digital elements, which helps eliminate errors that can arise during analogue computation. This forms part of the deep technology we have developed and do not disclose publicly.

Components such as ADCs and DACs require careful design to ensure accurate conversion between classical analogue signals and our DigAn architecture. Proper design of these components is essential to maintain the performance and accuracy of our AI processors.

Can you briefly explain the role of memory in your AI processors?

In the embedded world, small devices function as complete computers, storing all data, programs, and AI models on-device rather than relying on external memory or storage. When an application requires a larger AI model, it must fit entirely into on-chip memory.

Our first version offered 320 kilobytes of on-chip memory, which developers quickly found limiting. The new version expands this to 2 megabytes, enabling larger applications, more complex operating systems, additional peripherals and